

MIKHAYLOVKA, Kazakhstan

WMO: 298020

Lat: 53.8394N

Lon: 76.5411E

Elev: 374

StdP: 14.50

Time Zone: 6.00 (E06)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-31.3	-26.3	-35.9	0.7	-31.1	-31.0	1.0	-26.2	21.8	21.1	19.4	19.9	4.2	330	0.616

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.6	89.1	65.8	85.9	64.7	82.4	63.5	68.4	81.8	67.0	79.9	65.7	78.0	8.1	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.4	91.7	71.8	62.9	87.0	70.8	61.2	81.9	70.4	32.9	81.8	31.8	79.8	30.7	77.8	80.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
20.3	17.6	15.6	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-32.9	95.1	6.1	3.5	-37.3	97.6	-40.9	99.7	-44.4	101.7	-48.8	104.2
			-33.7	71.8	7.4	3.0	-39.0	73.9	-43.3	75.6	-47.5	77.3	-52.9	79.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	36.4	-1.2	2.5	20.0	41.5	54.4	66.1	67.9	64.1	53.1	39.0	21.4	5.5	
	DBStd	26.90	15.37	13.27	11.06	10.86	9.33	7.74	6.28	6.79	8.54	9.03	13.04	14.57	
	HDD50	6866	1586	1331	930	297	64	3	0	0	61	357	858	1379	
	HDD65	10797	2051	1751	1395	706	344	78	43	101	370	806	1308	1844	
	CDD50	1890	0	0	0	41	201	487	555	436	153	17	0	0	
	CDD65	345	0	0	0	0	17	112	132	72	12	0	0	0	
	CDH74	3844	0	0	0	24	296	1202	1293	835	192	2	0	0	
	CDH80	1256	0	0	0	3	73	412	440	272	56	0	0	0	
Wind	WSAvg	7.4	7.0	7.1	8.1	8.4	8.3	6.9	6.5	6.5	6.8	7.2	7.9	7.6	
Precipitation	PrecAvg	11.9	0.6	0.4	0.4	0.6	1.1	1.5	2.2	1.7	1.2	1.1	0.8	0.6	
	PrecMax	21.1	2.4	0.9	1.5	1.7	5.5	5.0	5.8	5.8	3.3	2.4	2.1	1.5	
	PrecMin	5.5	0.1	0.0	0.0	0.0	0.3	0.0	0.1	0.2	0.4	0.1	0.0	0.0	
	PrecStd	3.5	0.5	0.3	0.3	0.4	1.1	1.0	1.5	1.3	0.7	0.6	0.5	0.4	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.0	33.9	47.4	77.5	87.9	92.6	93.2	92.4	87.1	69.4	46.7	34.8	
		MCWB	32.2	32.0	40.4	57.8	61.7	66.1	69.0	67.4	62.5	52.7	42.6	33.0	
	2%	DB	29.6	31.6	40.1	70.6	81.2	88.2	88.6	87.0	79.4	63.3	42.5	32.9	
		MCWB	27.9	30.0	36.1	54.9	58.9	65.0	66.9	65.1	58.7	50.3	39.0	31.5	
	5%	DB	25.9	27.4	36.7	64.9	77.0	85.1	85.0	82.6	73.6	57.3	38.8	30.2	
		MCWB	24.5	26.0	33.9	51.4	57.5	64.1	65.7	63.4	56.8	47.0	36.0	28.8	
	10%	DB	22.1	22.7	34.5	58.9	72.0	81.1	81.2	78.6	68.1	52.2	35.8	26.6	
		MCWB	21.0	21.4	32.3	48.2	55.3	63.1	65.0	62.1	54.6	44.7	33.4	25.2	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	32.5	32.4	40.9	58.8	63.5	69.1	71.1	69.1	64.3	54.0	43.2	33.5
			MCDB	33.8	33.6	46.5	74.6	83.6	84.6	85.2	83.9	84.0	65.2	46.3	34.3
2%		WB	28.3	29.9	36.5	55.7	60.5	67.4	69.0	66.8	60.7	51.1	39.5	31.6	
		MCDB	29.6	31.5	39.6	67.7	77.1	82.5	81.5	80.0	74.9	61.9	42.3	32.7	
5%		WB	24.6	26.0	34.3	52.3	58.6	66.1	67.7	65.4	58.0	47.9	36.3	28.9	
		MCDB	26.0	27.4	36.4	63.5	73.8	79.6	79.8	78.2	71.4	55.5	38.4	30.1	
10%		WB	21.0	21.4	32.4	48.7	56.5	64.5	66.4	63.7	55.4	45.3	33.7	25.2	
		MCDB	22.1	22.7	34.5	57.7	69.9	77.8	77.9	76.1	66.5	51.6	35.5	26.6	
Mean Daily Temperature Range	5% DB	MDBR	13.7	14.5	14.9	19.1	22.4	21.0	19.6	20.9	19.5	15.4	12.1	14.1	
		MCDBR	13.6	12.7	13.2	28.0	29.1	25.9	25.4	28.0	28.4	24.8	11.4	14.8	
		MCWBR	13.2	12.3	10.7	14.7	13.0	9.6	9.0	11.1	12.6	13.7	9.0	14.7	
	5% WB	MCDBR	13.8	12.8	13.3	26.8	25.7	22.2	20.7	23.3	25.6	21.8	10.9	15.0	
		MCWBR	13.4	12.4	11.4	14.6	12.9	9.1	8.6	10.3	12.3	13.3	9.1	15.0	
Clear-Sky Solar Irradiance	taub	0.292	0.322	0.381	0.405	0.407	0.413	0.436	0.423	0.370	0.351	0.330	0.295		
	taud	2.039	1.993	1.929	2.155	2.208	2.274	2.213	2.218	2.319	2.256	2.112	2.004		
	Ebn at Noon	199	233	244	259	266	266	256	252	252	226	182	166		
	Edh at Noon	23	34	46	42	42	40	42	39	31	26	22	20		
All-Sky Solar Radiation	RadAvg	353	705	1199	1491	1842	1971	1779	1501	1081	563	271	229		
	RadStd	25	27	87	115	148	120	122	115	100	71	35	27		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (11 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page